

Answers — items 1 and 2

The key to the two items we did **in the room**. Items 3 and 4 are **PS3**; their solutions post to bCourses after the deadline, not here. The drawing was never the hard part — the prose under each ► is.

1 · The constitutive unit — what the stability argument does not use

$$p^* = \frac{\alpha_0}{\mu} = \frac{23.1}{0.0231} = 1000 \text{ molecules}$$

► **Name one thing you could change about the production curve that would leave the stability argument untouched.** Almost anything: make it fall with p , make it rise and saturate, put a bump in it. The argument only needs the gap $\alpha(p) - \mu p$ to go from **positive to negative** as p increases through the crossing. It never asks what $\alpha(p)$ is.

That is not a throwaway. It is why the same argument survives into two dimensions on Tuesday, where you cannot draw a single crossing at all and have to reason about the sign of a flow instead. A stability criterion that depends on the functional form is a criterion you have to re-derive for every circuit; this one you do not.

Worth noticing: the one thing that **would** break it is a production curve that crosses the removal line from below with a steeper slope. You will meet exactly that crossing in item 3 — work out for yourself what the sign of the gap does either side of it.

2 • Negative autoregulation — which curve moved

Fix the level. At the crossing, production equals removal:

$$\frac{\alpha}{1 + p^*/K} = \mu p^* = \alpha_0 \quad \implies \quad \alpha = \alpha_0 \left(1 + \frac{p^*}{K}\right)$$

With $K = p^*$ that is $\alpha = 2\alpha_0 = 46.2$ per minute — the **repression ratio**, and it is the promoter strength you have to order.

► **Which curve moved, and which did not.** The **production** curve moved twice: down, when repression was added, and back up, when α was raised to restore the target. The **removal** curve never moved at all — it is μp before and after, because negative autoregulation does not touch γ or the growth rate.

So the synthesis rate **at the crossing** is $\mu p^* = \alpha_0 = 23.1$ per minute, exactly what it was in item 1. At steady state, NAR is free.

It is not free overall, and that distinction is the session. The bill $\alpha = 46.2$ is what the promoter fires at while p is still small and the operator still empty — early in the transient, which is when the extra protein gets made. The lunch is not free; it is prepaid. PS3 asks you to price it over a day against the tag route, which pays every minute forever instead of once.

► **Why it is faster.** At $p = p^*/2 = 500$: item 1 has production **23.1** and removal **11.6**, a gap of **11.5**. Item 2 has production $46.2/1.5 = 30.8$ and the same removal **11.6**, a gap of **19.2** — **two-thirds larger**. The gap is dp/dt , so the NAR circuit is climbing harder at the same level, and it keeps that advantage all the way up because its production curve is falling toward the crossing while the flat one just waits there. Half-time **16.8** min against **30.0** — a **1.79-fold** speed-up. That is the same as the best degradation tag in Andersen 1998 buys (**1.75**), to within the precision of a measured half-life; do not read the 2% as a win. What matters is that NAR gets it **without moving the removal line at all**. The ratio that would match the tag exactly is **1.94**.

► **Let $K \rightarrow \infty$.** The repressor never occupies the operator, so the promoter never sees it; $\alpha/(1 + p/K) \rightarrow \alpha$, and the required α collapses to α_0 . A circuit with an operator nothing binds is a constitutive circuit — and notice you can say that without doing the limit.

► **Make it aggressive: $K = p^*/10$.** Then $\alpha = 11\alpha_0 = 254$ per minute: an eleven-fold stronger promoter, and a much faster circuit. But the synthesis rate at the crossing is **still $\mu p^* = 23.1$** . The steady-state cost did not move, because the removal curve did not move and the crossing is where the two agree.

What changed and what did not is the whole result. Speed came from the *shape* of the production curve; cost is set by the *crossing*, and the crossing is pinned by removal. That is why a degradation tag — which moves the removal line — is the expensive route and this one is not.

The honest caveat: aggressive repression is not free in practice. An eleven-fold promoter is eleven-fold leaky before the repressor accumulates, the transient overshoot is larger, and you have spent a strong promoter and a tight operator that you might have wanted for something else.